

## BANKING SQL ASSIGNMENT (INTERMEDIATE)

### 1. Setup (Given Schema)

```
-- =====  
  
-- BANKING DATABASE SETUP  
  
-- =====  
  
DROP DATABASE IF EXISTS banking_lab;  
CREATE DATABASE banking_lab;  
USE banking_lab;  
  
-- Customers  
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_name VARCHAR(100),  
    city VARCHAR(50)  
);  
  
-- Accounts  
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY AUTO_INCREMENT,  
    customer_id INT,  
    account_type VARCHAR(50), -- Savings / Current  
    balance DECIMAL(12,2),  
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
);  
  
-- Transactions  
CREATE TABLE transactions (
```

```
transaction_id INT PRIMARY KEY AUTO_INCREMENT,  
account_id INT,  
transaction_type VARCHAR(50), -- Deposit / Withdrawal  
amount DECIMAL(12,2),  
transaction_date DATE,  
FOREIGN KEY (account_id) REFERENCES accounts(account_id)  
);
```

## 2. Sample Data

-- Customers

```
INSERT INTO customers (customer_name, city) VALUES  
( 'Aung Aung', 'Yangon'),  
( 'Su Su', 'Mandalay'),  
( 'Kyaw Kyaw', 'Yangon'),  
( 'Mya Mya', 'Bago'),  
( 'Ko Ko', 'Yangon');
```

-- Accounts

```
INSERT INTO accounts (customer_id, account_type, balance) VALUES  
(1, 'Savings', 5000),  
(2, 'Current', 8000),  
(3, 'Savings', 12000),  
(4, 'Savings', 3000),  
(5, 'Current', 15000);
```

-- Transactions

```
INSERT INTO transactions (account_id, transaction_type, amount, transaction_date) VALUES  
(1, 'Deposit', 2000, '2025-01-01'),  
(1, 'Withdrawal', 500, '2025-01-02'),
```

(2, 'Deposit', 3000, '2025-01-03'),  
(3, 'Deposit', 4000, '2025-01-04'),  
(3, 'Withdrawal', 1000, '2025-01-05'),  
(4, 'Deposit', 1500, '2025-01-06'),  
(5, 'Deposit', 7000, '2025-01-07'),  
(5, 'Withdrawal', 2000, '2025-01-08');



### 3. Assignment Questions

#### ◆ PART A: SELECT & FROM

1. Display all customers
2. Show customer name and city only
3. Display all accounts with balance

#### ◆ PART B: WHERE

4. Find customers from **Yangon**
5. Find accounts with balance greater than 10,000
6. Show all **Deposit transactions only**
7. Find transactions between '2025-01-01' and '2025-01-05'

#### ◆ PART C: ORDER BY

8. Display accounts sorted by balance (highest first)
9. Show transactions sorted by date (latest first)

#### ◆ PART D: GROUP BY

10. Total balance per account type
11. Total transaction amount per account
12. Number of transactions per account

◆ **PART E: HAVING**

13. Show account types with total balance > 10,000
14. Show accounts with total transaction amount > 5,000

◆ **PART F: COMBINED QUERY**

15. For each customer:
  - Show total transactions
  - Show total amount
  - Only include customers with total amount > 3,000
  - Sort by total amount descending

◆ **PART G: ADVANCED (JOIN + CLAUSES)**

16. Show customer name, account type, and balance
17. Show total deposits per customer
18. Show customers who made more than 1 transaction
19. Show customers with total withdrawals > 1,000